



FREE OPERATOR GUIDE

The Complete Guide to Recipe Costing

Batch recipes, unit conversions, and the hidden math behind true cost per plate.

Built for restaurant owners, chefs, breweries, caterers, and food-service operators who want to know their true cost per plate down to the penny.

Use this guide to cost raw ingredients, nested batch recipes, unit mismatches, yield loss, and spice conversions with confidence.

MenuForge.ai | Recipe management, costing, and AI-powered menu operations

SECTION 1

Why Recipe Costing Matters

Every dish you serve has a cost. Not just the obvious protein or produce, but every pinch of salt, every splash of oil, every batch of sauce that took an hour to prepare.

Most kitchens estimate these costs loosely and hope margins work out. Hope is not a financial strategy.

ACCURATE RECIPE COSTING LETS YOU

- Set menu prices that protect your margins.
- Spot which dishes are profit drivers and which are quietly losing money.
- Negotiate with suppliers using real usage data.
- Adjust portions or substitutions with confidence.

The challenge is that real-world recipes are not flat ingredient lists. They contain sub-recipes, unit mismatches, and yield losses that make a simple spreadsheet unreliable. This guide walks you through a repeatable method for handling all of them.

SECTION 2

The Basics: Flat Recipe Costing

Start with the simplest case: a recipe where every ingredient is a raw purchased item.

$$\text{Cost per Serving} = \text{Sum of (Ingredient Qty x Unit Cost)} / \text{Number of Servings}$$

Example: House Vinaigrette

House Vinaigrette makes 32 oz, or 16 servings of 2 oz each.

Ingredient	Qty Used	Purchase Price	Purchase Size	Line Cost
Olive Oil	24 oz	\$12.00	128 oz (1 gal)	\$2.25
Red Wine Vinegar	6 oz	\$4.50	32 oz	\$0.84
Dijon Mustard	1 oz	\$5.00	16 oz	\$0.31
Honey	0.5 oz	\$8.00	24 oz	\$0.17
Salt	0.25 oz	\$2.50	26 oz	\$0.02
Black Pepper	0.1 oz	\$6.00	16 oz	\$0.04
Total Batch Cost				\$3.63
Cost per 2-oz Serving				\$0.23

The formula for each line is straightforward: (Qty Used / Purchase Size) x Purchase Price. The real complexity begins when one of those "ingredients" is itself a recipe.

SECTION 3

Recipe Inception: Batch Recipes as Ingredients

In a professional kitchen, many ingredients are prepared in-house: stocks, sauces, spice blends, doughs, brines. These batch recipes are then used as ingredients in other recipes, and those recipes may themselves be used in yet another recipe.

We call this recipe inception: recipes nested inside recipes, sometimes several levels deep.

How Costs Propagate

- 1. **Level 0 - Raw ingredients:** These have a known purchase price and purchase unit. No calculation needed beyond the line-cost formula above.
- 2. **Level 1 - Batch recipes that use only raw ingredients:** Cost the entire batch, then divide by yield to get a cost per unit of output.
- 3. **Level 2 - Recipes that use Level 1 batches:** When a Level 1 batch appears as an ingredient, use its computed cost per unit as if it were a purchase price.
- 4. **Level N - Repeat as needed.**

KEY RULE

You must fully resolve a batch recipe's cost before you can cost any recipe that uses it. Work from the leaves of the dependency tree toward the root.

Worked Example: Marinara Sauce

Ingredient	Qty	Purchase Price	Purchase Size	Line Cost
Crushed Tomatoes	96 oz	\$4.50	96 oz (#10 can)	\$4.50
Olive Oil	4 oz	\$12.00	128 oz	\$0.38
Garlic, minced	2 oz	\$6.00	32 oz	\$0.38
Dried Oregano	0.5 oz	\$8.00	8 oz	\$0.50
Salt	0.5 oz	\$2.50	26 oz	\$0.05
Sugar	1 oz	\$3.50	80 oz	\$0.04
Total Batch Cost				\$5.85

Marinara cost per oz = \$5.85 / 128 oz = \$0.0457/oz.

Use the Batch in a Parent Recipe: Chicken Parmesan

Ingredient	Qty	Unit Cost	Source	Line Cost
Chicken Breast	8 oz	\$0.0394/oz	Raw	\$0.32
Marinara Sauce	4 oz	\$0.0457/oz	Batch recipe	\$0.18
Mozzarella	2 oz	\$0.0313/oz	Raw	\$0.06
Breadcrumbs	1.5 oz	\$0.0188/oz	Raw	\$0.03
Egg	1 ea	\$0.25/ea	Raw	\$0.25
Total per Serving				\$0.84

Notice how the Marinara row uses the computed cost per ounce from Step 1, not a purchase price. The costing engine treats it exactly like a raw ingredient, but the unit cost is derived, not purchased.

SECTION 4

The Unit Conversion Problem

Perhaps the trickiest day-to-day costing challenge is unit mismatch: you buy an ingredient in one unit but use it in a completely different one.

The most common case is spices purchased by weight, measured in recipes by volume.

Why You Can't Just Equate Teaspoons to Grams

A teaspoon of ground cinnamon weighs about 2.6 grams. A teaspoon of garlic powder weighs about 3.1 grams. A teaspoon of paprika weighs about 2.3 grams. The density of each spice is different, so there is no single conversion factor from volume to weight.

The Density-Bridge Method

To convert, you need a density bridge: the weight-per-volume measurement for that specific ingredient.

$$\text{Line Cost} = (\text{Volume Used} \times \text{Density}) / \text{Purchase Weight} \times \text{Purchase Price}$$

Or equivalently: convert the recipe quantity to weight using density, then compute the cost as if it were a weight-based ingredient.

EXAMPLE: GROUND CUMIN

1. You buy ground cumin at \$14.00 per pound (453.6 g).
2. Your recipe uses 2 tsp.
3. Look up density: ground cumin is approximately 2.7 g/tsp.
4. Convert recipe quantity: $2 \text{ tsp} \times 2.7 \text{ g/tsp} = 5.4 \text{ g}$.
5. Compute cost: $(5.4 \text{ g} / 453.6 \text{ g}) \times \$14.00 = \$0.17$.

Common Spice Density Reference

These are averages. Actual density varies by grind, brand, and settling. For high-accuracy costing, weigh a measured teaspoon on a gram scale in your own kitchen.

Spice / Seasoning	g/ tsp	Spice / Seasoning	g/ tsp
Allspice, ground	2.1	Mustard, ground	2.3
Basil, dried	0.7	Nutmeg, ground	2.2
Black Pepper, ground	2.3	Onion Powder	2.8
Cayenne Pepper	1.8	Oregano, dried	1.0
Chili Powder	2.6	Paprika	2.3
Cinnamon, ground	2.6	Red Pepper Flakes	1.5
Coriander, ground	1.8	Rosemary, dried	1.2
Cumin, ground	2.7	Sage, ground	0.7
Curry Powder	2.0	Salt, fine	6.0
Dill Weed, dried	1.0	Salt, kosher (Morton)	4.8
Garlic Powder	3.1	Smoked Paprika	2.3
Ginger, ground	1.8	Thyme, dried	1.4
Italian Seasoning	1.0	Turmeric, ground	3.0

PRO TIP

Establish a density reference sheet for every spice in your pantry. Weigh a level teaspoon once, record it, and use that number in all future costings.

SECTION 5

Other Common Unit Mismatches

Liquids bought by volume, used by weight: Use the liquid's specific gravity. Olive oil is about 0.92 g/ mL; honey is about 1.42 g/mL.

"Each" items with weight-based pricing: Eggs bought by the dozen but priced per pound require an average egg weight so you can derive cost per each.

Prep yield loss: If you buy whole carrots at \$1.20/ lb but only 85% is usable after peeling, your effective cost per pound of usable carrot is $\$1.20 / 0.85 = \$1.41/\text{lb}$.

Putting It All Together

1. **Map the dependency tree.** List every sub-recipe used by your target recipe, and any sub-recipes they use. Draw it out if it helps.
2. **Resolve from the bottom up.** Start with batch recipes that contain only raw ingredients. Cost them fully. Record a cost-per-unit-of-output for each.
3. **Reconcile units at every line.** For each ingredient row, make sure the recipe quantity and purchase or batch-output unit are compatible. If not, apply a density bridge or yield factor.
4. **Compute line costs, then sum.** Each line = Qty Used x Cost Per Unit after conversion. Sum all lines for the total batch cost, then divide by servings or yield.
5. **Review and maintain.** Costs change with supplier pricing, seasonal availability, and portion drift. Re-cost at least quarterly or whenever a key price changes significantly.

COMMON PITFALL

Forgetting to update a batch recipe's cost when one of its ingredients changes price. In a nested system, a price change at the leaf level must cascade up through every recipe that depends on it. Manual spreadsheets almost always miss some of these cascades.



Let MenuForge Do the Math

MenuForge automates everything in this guide: batch recipe inception, automatic unit conversion, cost cascading when prices change, and real-time cost-per-serving for every dish on your menu.

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